Two-layer perceptron from scratch

1 Inputs

$$\mathbf{X} \in \mathbb{R}^{N \times D}, \mathbf{y} \in \mathbb{R}^{N \times C}$$

2 The Network

$$\begin{aligned}
\mathbf{W}_1 \in \mathbb{R}^{D \times H}, \mathbf{b}_1 \in \mathbb{R}^{1 \times H} \\
\mathbf{W}_2 \in \mathbb{R}^{H \times C}, \mathbf{b}_2 \in \mathbb{R}^{1 \times C}
\end{aligned}$$

$$\begin{aligned}
\mathbf{h}_1 = \mathbf{X}\mathbf{W}_1 + \mathbf{b}_1 \in \mathbb{R}^{N \times H} \\
\mathbf{h}_2 = f(\mathbf{h}_1) \in \mathbb{R}^{N \times H} \\
\mathbf{h}_3 = \mathbf{h}_2\mathbf{W}_2 + \mathbf{b}_2 \in \mathbb{R}^{N \times C}
\end{aligned}$$

Note: Below, we will use the notations $\mathbf{h}_i$ and $\mathbf{h}^{(i)}$ interchangeably so as the make the text more legible. In particular, when it's necessary to index into $\mathbf{h}_i$, we will use the $\mathbf{h}_j^{(i)}$ notation to refer to the jth element in $\mathbf{h}_i$.

3 Nonlinearity

$$f(\mathbf{X}) = \mathrm{ReLU}(\mathbf{X}) = \begin{bmatrix} \mathrm{ReLU}(x_{11}) & \cdots & \mathrm{ReLU}(x_{1j}) & \cdots & \mathrm{ReLU}(x_{1n}) \\ \vdots & \ddots & & & \vdots \\ & & \mathrm{ReLU}(x_{ij}) & & \\ \vdots & & & \ddots & \vdots \\ \mathrm{ReLU}(x_{m1}) & \cdots & \mathrm{ReLU}(x_{mj}) & \cdots & \mathrm{ReLU}(x_{mn}) \end{bmatrix}$$

$$= \begin{cases} x_{ij} & \text{, if } x_{ij} \geq 0 \\ 0 & \text{, if } x_{ij} < 0 \end{cases}$$

4 Predictions

$$
\begin{aligned}
\mathbf{q}_i &= \mathrm{softmax}(\mathbf{h}_3)_i \\
\mathrm{softmax}(\mathbf{z})_i = \mathrm{softmax}(\mathbf{z}_i) &= \frac{\mathrm{e}^{\mathbf{z}_i}}{\sum_{k=1}^{C} \mathrm{e}^{\mathbf{z}_k}}
\end{aligned}
$$

5 Loss

$$
\begin{aligned}
\mathcal{L} = \mathrm{CE}(\underbrace{p}_{\text{true}}, \underbrace{q}_{\text{estimate}}) &= -\mathbb{E}_p[\log q] \\
&= -\sum_{k=1}^{C} \mathbf{p}_k \log \mathbf{q}_k \\
&= -\cancelto{1}{\mathbf{p}_i} \cdot \log\left(\frac{\mathrm{e}^{\mathbf{h}_i^{(3)}}}{\sum_{k=1}^{C} \mathrm{e}^{\mathbf{h}_k^{(3)}}}\right) \\
&= -\log \mathrm{e}^{\mathbf{h}_i^{(3)}} + \log\left(\sum_{k=1}^{C} \mathrm{e}^{\mathbf{z}_k}\right) \\
&= \boxed{\log\left(\sum_{k=1}^{C} \mathrm{e}^{\mathbf{z}_k}\right) - \mathbf{h}_i^{(3)}}
\end{aligned}
$$

6 Gradients (backprop)

Outputs:

$$
\nabla_{\mathbf{h}_i^{(3)}} \mathcal{L}_i = \begin{cases} \cancelto{\mathbf{q}_i}{\frac{1}{\sum_k e^{\mathbf{h}_k^{(3)}}} \cdot \mathrm{e}^{\mathbf{h}_i^{(3)}}} - 1 & , i \text{ is correct class} \\ \mathbf{q}_i + 0 & , i \text{ is incorrect class} \end{cases}
$$

$$
\boxed{\nabla_{\mathbf{h}_i^{(3)}} \mathcal{L} = \mathbf{q} - \mathbf{p}} \in \mathbb{R}^{N \times C}
$$

Layer 2:

$$\frac{\partial \mathbf{h}_3}{\partial \mathbf{W}_2} = \mathbf{h}_2^\top \Rightarrow \frac{\partial \mathcal{L}}{\partial \mathbf{W}_2} = \frac{\partial \mathbf{h}_3}{\partial \mathbf{W}_2} \cdot \frac{\partial \mathcal{L}}{\partial \mathbf{h}_3}$$

$$\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{W}_2} = \mathbf{h}_2^\top(\mathbf{q} - \mathbf{p})} \in \mathbb{R}^{H \times C}$$

$$\frac{\partial \mathbf{h}_3}{\partial \mathbf{b}_2} = 1 \Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{b}_2} = \mathbf{q} - \mathbf{p}} \in \mathbb{R}^{N \times C}$$

$$\frac{\partial \mathbf{h}_3}{\partial \mathbf{h}_2} = \mathbf{W}_2^\top \Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_2} = (\mathbf{q} - \mathbf{p})\mathbf{W}_2^\top} \in \mathbb{R}^{N \times H}$$

Layer 1 nonlinearity:

$$\frac{\partial \mathbf{h}_2}{\partial \mathbf{h}_1} = \nabla \mathrm{ReLU} = \begin{cases} +1 & \mathbf{h}_{ij}^{(1)} \geq 0 \\ 0 & \mathbf{h}_{ij}^{(1)} < 0 \end{cases} = \mathbb{1}(\mathbf{h}_{ij}^{(1)})$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \mathbf{h}_1} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_2} \frac{\partial \mathbf{h}_2}{\partial \mathbf{h}_1}$$

$$\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_1} = \mathbb{1}(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_2})} \in \mathbb{R}^{N \times H}$$

Layer 1:

$$\frac{\partial \mathbf{h}_1}{\partial \mathbf{W}_1} = \mathbf{X}^\top$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial \mathbf{W}_1} = \frac{\partial \mathbf{h}_1}{\partial \mathbf{W}_1} \frac{\partial \mathcal{L}}{\partial \mathbf{h}_1}$$

$$\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{W}_1} = \mathbf{X}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{h}_1}} \in \mathbb{R}^{D \times H}$$

$$\frac{\partial \mathbf{h}_1}{\partial \mathbf{b}_1} = 1$$

$$\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{b}_1} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_1}} \in \mathbb{R}^{N \times H}$$